

Algebra 3 — Quiz 2

Time: 1 hour 15 minutes

1. Let R be a unique factorization domain, and let $a, b \in R$ be nonzero. Suppose that

$$a^m = b^n$$

for relatively prime positive integers m and n .

Prove that there exist $c \in R$ and units $u, v \in R^\times$ such that

$$a = uc^n, \quad b = vc^m.$$

What relation must hold between u and v ?

2. Let R be a unique factorization domain and let K be its field of fractions. Suppose that $\alpha \in K$ satisfies

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0, \quad a_i \in R.$$

Prove that $\alpha \in R$.

3. Let

$$R = \mathbb{Z}[\sqrt{2}].$$

For $\alpha = a + b\sqrt{2} \in R$, define

$$N(\alpha) = |a^2 - 2b^2|.$$

Prove that R is a Euclidean domain with respect to N . Thus, for every $\alpha, \beta \in R$ with $\beta \neq 0$, prove that there exist $q, r \in R$ such that

$$\alpha = q\beta + r,$$

where either $r = 0$ or $N(r) < N(\beta)$.

4. Determine all integer solutions $(x, y) \in \mathbb{Z}^2$ to

$$x^2 + 1 = y^3.$$

You may use that $\mathbb{Z}[i]$ is a unique factorization domain.

5. Let x, y, z be positive integers satisfying

$$x^2 + y^2 = z^2, \quad \gcd(x, y) = 1,$$

and suppose that y is even.

Using unique factorization in $\mathbb{Z}[i]$, prove that there exist relatively prime positive integers $m > n$ of opposite parity such that

$$x = m^2 - n^2, \quad y = 2mn, \quad z = m^2 + n^2.$$

You should justify the following steps rather than quote the parametrization of primitive Pythagorean triples:

(a) Show that $x + iy$ and $x - iy$ have no common nonunit divisor in $\mathbb{Z}[i]$.

(b) Use

$$(x + iy)(x - iy) = z^2$$

and unique factorization to show that $x + iy$ is a unit times a square in $\mathbb{Z}[i]$.

(c) Deal carefully with the possible units in $\mathbb{Z}[i]$, and deduce the stated formulas and the parity and coprimality conditions on m, n .